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EXPERIMENT – 6 –Work with APIs and fetch data asynchronously using JavaScript.

AIM:
Work with APIs and fetch data asynchronously using JavaScript.

PROCEDURE:

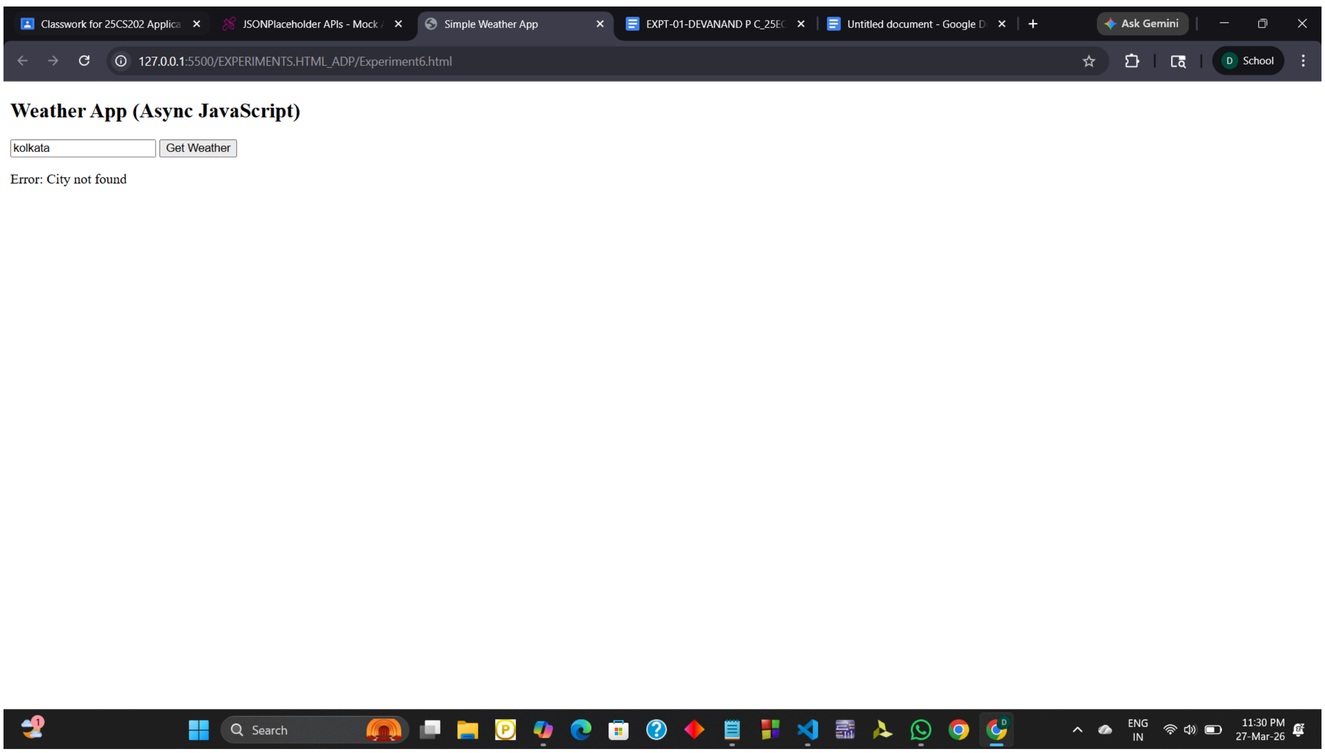
1. Open Visual Studio Code and create a new HTML file.
2. Write the basic HTML structure using html tags.
3. Add input field to accept the city name from the user.
4. Add a button to trigger the weather fetching function.
5. Use JavaScript async and await functions to fetch data from OpenWeatherMap API.
6. Construct the API URL with city name and API key.
7. Use the fetch() method to send a request and receive a response.
8. Convert the response into JSON format using the .json() method.
9. Display temperature and weather description on the webpage.
10. Handle errors using try-catch blocks.
11. Run the program in the browser and test with different city names.

PROGRAM CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Weather App</title>
</head>
<body>

  <h2>Weather App (Async JavaScript)</h2>

  <input type="text" id="city" placeholder="Enter city name">
  <button onclick="getWeather()">Get Weather</button>

  <p id="result"></p>

  <script>
    async function getWeather() {
      let city = document.getElementById("city").value;

      let apiKey = "YOUR_API_KEY";
      let url =
`https://api.openweathermap.org/data/2.5/weather?q=${city}&appid=${apiKey}&units=metric`;

      try {
        let response = await fetch(url);

        if (!response.ok) {
          throw new Error("City not found");
        }

        let data = await response.json();

        let temp = data.main.temp;
        let weather = data.weather[0].description;

        document.getElementById("result").innerHTML =
          "Temperature: " + temp + " °C <br> Weather: " + weather;
      }
      catch(error) {
```

```
                document.getElementById("result").innerHTML = "Error: " +
error.message;
            }
        }
    </script>

</body>
</html>
```

OUTPUT:



RESULT:

A simple weather web application was successfully created using HTML and asynchronous JavaScript. The application fetches real-time weather data from an API and displays temperature and weather conditions dynamically. The program handles errors effectively and provides accurate results for different city inputs.